Алгебра. 30.11

Функ-е опр. рац. дробей: $\frac{f}{g}$ (дробно-рац. функции)

Т. 1 $\frac{f}{g} = \frac{q}{r} \wedge (f,g) = 1 \wedge (q,r) = 1$, т.е. обе дроби взаимно простые

$f, g, q, r \in K[x]$

$$\Downarrow$$

$$(\exists c \in P) \quad cf = q \wedge cg = r$$

Док-во: 1) $\frac{fr}{g} = q \Rightarrow g \mid r \Rightarrow \deg g \le \deg r$.

$f = \frac{qg}{r} \Rightarrow r \mid g \Rightarrow \deg r \le \deg g$.

$\Rightarrow \frac{f}{g} = \frac{q}{cg} \; |\cdot g \Rightarrow f = \frac{q}{c} \Leftrightarrow cf = q$.

Опр.: $\frac{f}{g}$ — правильная $\Leftrightarrow \deg f < \deg g$.

Т. 2 $\frac{f}{g} = p + \frac{r'}{g'} \wedge (r', g') = 1 \wedge (\deg r' < \deg g')$

Док-во: (2) $\deg f \ge \deg g$

т.е. можно разделить:

(3) $f = gQ + r$

$\deg r < \deg g$

(4) $\frac{f}{g} \overset{||}{=} Q + \frac{r}{g} = Q + \frac{r'}{g'}$.

От противного: $\frac{f}{g} = Q' + \frac{r''}{g''} \; |\cdot g$

$$f = Q'g + \frac{r''}{g''}g$$

$$\deg \frac{r''}{g''}g = \deg r'' + \underbrace{\deg g - \deg g''}_{< \deg g}$$

$$Q = Q' \wedge r = \frac{r''}{g''}g \;\Big|\; \cdot \tfrac{1}{g}$$

$$\Downarrow$$

$$\underbrace{\frac{r'}{g'} = \frac{r}{g} = \frac{r''}{g''}}_{\text{отличаются на пост. множ.}}$$

Т.3 $F = p_1^{k_1} \ldots p_m^{k_m}$; $\deg F = \sum_{i=1}^{n} k_i$.

Док-во: $(6) \Rightarrow \underbrace{F_0 F_0'}_{F''} \wedge \deg F_0 \geq 1 \wedge \deg F_0' \geq 1$

$F_0 = F_1 F_1' \quad \deg F_1, F_1' \geq 1.$

$$\Downarrow$$

$F_0 F_1 F_1' \Rightarrow \ldots \Rightarrow F_0 F_1 \ldots F_n$

приводим

Т. 4 $f, \hat{f}$ — неприв. м-мы $\Leftrightarrow (\nexists c \in K) cf = \hat{f} \Rightarrow (f, \hat{f}) = 1$.

Док-во: — от противного:

$\exists (f, \hat{f}) = d, \deg d \geq 1 \Rightarrow$

$d \mid f \qquad\qquad d \mid \hat{f}$

$\Downarrow \qquad\qquad \Downarrow$

$\underbrace{\exists c : d = cf \qquad \exists c' : d = c'\hat{f}}$

$cf = c'\hat{f} \Leftrightarrow \frac{c}{c'} = 1 = \hat{f}$

Опр.: Почти простейшая дробь: $\frac{p}{f^r}$, $\deg p < f^r \wedge f$ — неприв.

Т. 5.

$$(6) \Rightarrow (p_m^{k_m}, p_1^{k_1}, \dots, p_{m-1}^{k_{m-1}}) = 1.$$

вз.-прост. с произв.-м

Т. 6

$$\frac{f}{g} = Q + \frac{r'}{g'}, \quad \deg r' < \deg g' \Rightarrow \frac{r'}{g'}$$ представима как Σ п.п. дробей

$$g' \neq \gamma^k, \quad k \geq 1.$$

$$\Downarrow$$

$$\exists \gamma, \exists_{k \geq 1} : \gamma^k \mid g' \wedge \neg(\gamma^{k+1} \mid g')$$

$$\left(\gamma^k, \frac{g'}{\gamma^k}\right) = 1.$$

$$\Downarrow$$

$$(\exists A, B \in k[x]) \; 1 = A\gamma^k + B\frac{g'}{\gamma^k} \; \Big| \cdot f'$$

$$f' = f'A\gamma^k + f'B\frac{g'}{\gamma^k} \; \Big| : g' \Rightarrow \frac{f'}{g'} = \frac{f'A}{\frac{g'}{\gamma^k}} + \underbrace{\frac{f'B}{\gamma^k}}_{= Q_1 + \frac{B'}{\gamma^k}}$$

Опр: $\frac{f}{\gamma^k}$ $\quad \deg f < \deg \gamma$

в $\mathbb{C}$: $f = \text{const } A$, $f = ax + b$

f — неприводимые

в $\mathbb{R}$: $f = ax + b$, $f = ax^2 + bx + c$, $4ac > b^2$

f — непр.

Т. 7 $\frac{F}{\gamma^k} \wedge \deg F < \deg \gamma^k \Rightarrow$

$$\Rightarrow (\exists F_i \quad) : \frac{F}{\gamma^k} = \sum_{i=1}^{k} \frac{F_i}{\gamma^i} \wedge \deg F_i < \deg \gamma$$

1. $\int \frac{dx}{ax-b} = \frac{1}{a} \ln\left(x + \frac{b}{a}\right),$

2. $\int \frac{dx}{(ax+b)^k} = \frac{1-k}{(ax-b)^{k-1}}$

3. $\int \frac{dx}{ax^2 + bx + c} = \frac{2}{\sqrt{4ac - a^2}} = \operatorname{arctg} \frac{2a + b}{\sqrt{4ac - b^2}}.$

$$4. \quad \int \frac{(ux+v)\,dx}{ax^2+bx+c} = \int \frac{(2ax+c)\frac{u}{2a}+\left(v-\frac{bu}{2a}\right)}{ax^2+bx+c}\,dx =$$

$$= \frac{u}{2a}\ln\left|ax^2+bx+c\right| + (\underset{\sim}{3}).$$